

Signature: Jessica AS-2Do not
write in
this
margin

1.a) 24

30

b) 1-2-3-4-go

c) odd
even

d) A method that returns void ~~can be used~~ can only be used as a procedure without a special value returned. ~~In (b), it can be used~~ As shown in (b), it can be used directly in the main program.
A method that returns a value can't be directly used in a main program but like a variable in the main program like in (c).

2a) ~~from 1 to 5~~ [0,5]

(i) [1,6]

(ii) Because the printing order runs for four times.

(iv) It can be easier to maintain. People will know ~~to~~ which is the line used to do the random clearly in this program.

It is convenient to use. In this program, it is used directly.

b) `public static int roll (int) {`

```

    return rnd.nextInt(6)+1;
}

```

```

(ii) public static void main(String[] args) {
    for (int i = 0; i < 5; i++) {
        System.out.print(roll() + " ");
    }
}

```

(i) ~~10~~ From 0 to 1.
circle

ii) cnt is the variable for counting the number of points located in the circle. N is the total number of points. The area of the square is 1 with a side length 1, the area of the circle is πr^2 , with radius 0.25, so its area is 0.25π . So the new variable P_i equals to ~~r^2 or r^2 divided by~~ 1 divided by 0.25 multiply by cnt/N, which is $4.0 * cnt / N$.

iii) 4.0 is a double type, while the type of 4 is integer. cnt is a double type, so the types need to be same.

```

3.a) FUNCTION nextTerm(n: INTEGER) RETURNS INTEGER
    IF n MOD 2 = 0 THEN
        RETURN n DIV 2
    ELSE
        RETURN 3*n+1
    ENDIF
ENDFUNCTION

```

```

b) FUNCTION steps(num num: INTEGER) RETURNS INTEGER
    DECLARE cnt: INTEGER
    cnt <- 0

```

~~REPEAT~~

~~IF~~ ~~nextTerm(num)~~

WHILE num < > 1 DO

num $\leftarrow$ nextTerm(num)

cnt $\leftarrow$ cnt + 1

ENDWHILE

RETURN cnt

ENDFUNCTION.

c) ~~Because by using a counter loop, we can~~
Because we don't know the exact times of loops need
to run Using a pre-condition loop is proper

d) DECLARE ~~num~~ numb : INTEGER

INPUT numb

DECLARE index : INTEGER

FOR index $\leftarrow$ 1 TO steps(numb)

OUTPUT numb $\leftarrow$ OUTPUT " "

numb $\leftarrow$ nextTerm(numb)

~~next~~ index

OUTPUT numb

OUTPUT "Steps:", steps(numb)